

The CMAGIC Cookbook

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Contents

The CMAGIC Cookbook	1
1. The physical hypothesis	2
2. The color–magnitude diagram and the linear region	2
3. The three measurement modes	3
4. K-corrections (the WS06 scheme)	4
5. Extinction	4
6. Standardization and the distance	5
7. Failure semantics and provenance (what makes this pipeline safe to use blind)	5
References	6
Part II — CMAGIC at High Redshift	6
8. The cross-filter problem and the template-synthesis solution	6
9. Sparse light curves: the fixed-slope estimator with full covariance	7
10. Validation and application scope	7

The CMAGIC Cookbook

Color–MAGnitude Intercept Calibration: Type Ia supernova distances from the magnitude at a fixed color.

Contents — Part I (low redshift): §1 the hypothesis; §2 the linear region; §3 modes L/Q/R; §4 K-corrections; §5 extinction; §6 standardization; §7 failure gates and provenance. Part II (high redshift): §8 cross-filter template synthesis; §8.1 the iterative template fit and the insensitivity of B_BV0.6; §9 Mode S (sparse, full covariance); §10 validation (SDSS-II) and application scope; §10.1 the SDSS-II worked example (numbers).

This document describes the method implemented by this package, in enough detail to use it critically. It is based on the two founding papers — Wang, Goldhaber, Aldering & Perlmutter (2003, ApJ 590, 944; hereafter W03) and Wang, Strovink, et al. (2006, ApJ 641, 50; hereafter WS06) — together with a reformulation (three measurement modes and explicit failure gates) developed for archival applications. Users who only want a distance can read §1 and then the README; the pipeline makes every choice described here automatically and reports what it chose.

1. The physical hypothesis

After maximum light, a Type Ia supernova's B-band magnitude and its B–V color evolve *together*: on the color–magnitude diagram (B versus B–V), the supernova travels a remarkably straight line for much of the first month past maximum, as the ejecta expand and cool. The CMAGIC hypothesis is:

The B magnitude at a fixed color, $B-V = 0.6$ mag — written $B_{BV0.6}$ — is a calibratable standard candle, in exactly the sense that the peak magnitude is the standard candle of light-curve fitters such as SALT3.

Why measure brightness at fixed *color* rather than at fixed *time* (peak)?

1. The locus is straight and slowly traversed. The linear color–magnitude relation (slope $\beta_{BV} = 1.94 \pm 0.16$ after K-corrections; W03) is measured from many nights of data, not from the single moment of peak, so $B_{BV0.6}$ is statistically robust and does not require dense coverage of maximum light.
2. Reduced extinction sensitivity. Dust moves a supernova *along* a vector of slope $R_B \approx 4.1$ in the same diagram. The extinction correction to $B_{BV0.6}$ is therefore $(R_B - \beta_{BV}) \cdot E(B-V) \approx 2.2 \cdot E$ rather than $R_B \cdot E \approx 4.1 \cdot E$ — roughly half the sensitivity of a peak-magnitude method to reddening errors (W03).
3. Smaller intrinsic dispersion. In the low-extinction calibration sample, $B_{BV0.6}$ standardizes to $\sigma \approx 0.07$ – 0.10 mag, versus ≈ 0.10 mag for $B_{\max}$ (W03, Table 2).
4. Independence. The residuals of CMAGIC and peak-magnitude distances correlate only weakly (Pearson ≈ 0.15 ; WS06), so the two are usefully independent distance estimators from the same photometry.

The distance chain is then conceptually identical to any standard-candle method:

$B_{BV0.6} \rightarrow \text{K-correction} \rightarrow \text{extinction correction} \rightarrow M_{BV}(\Delta m_{15}) \text{ calibration} \rightarrow \mu, D$

2. The color–magnitude diagram and the linear region

Plot nightly B against nightly B–V. The locus has four parts:

- a pre-maximum branch (distinct; hysteresis relative to the post-max branch),
- a post-maximum shoulder in the first days after B maximum (curved; W03 exclude a two-day gap around the transition),
- the linear region — the CMAGIC branch proper, typically spanning $B-V \approx 0.2$ – 1.0 for an unreddened object,
- the color turnback, when B–V reaches its maximum ($t \approx +30$ d for normal decliners) and turns blueward; beyond it the relation is no longer the same line.

The fiducial window is decline-rate-scaled (WS06):

subtype	window (days past B max)
normal	[$7/\Delta m_{15}$, $30/\Delta m_{15}$]
91T-like	[$10/\Delta m_{15}$, $29/\Delta m_{15}$]

For $\Delta m_{15} = 1.1$ this is $+6.4 \dots +27.3$ d. The package intersects this window with an object-specific safety

exit at $t(B-V \text{ max}) - 3 \text{ d}$, computed from the object's own smoothed color curve, which protects objects with unusually early or late color maxima. Within the window, nights are combined by weighted averaging; no residual-based point rejection is applied beyond a conservative 3σ photometric clip.

Two practical caveats established on archival data:

- The window inherits Δm_{15} quality. A poorly measured decline rate (e.g., a light-curve fit with the stretch parameter at its bound) produces a nonsense window; the pipeline gates on Δm_{15} provenance.
- Slow decliners. For $\Delta m_{15} \lesssim 0.9$ the $7/\Delta m_{15}$ entry can admit the tail of the post-max shoulder, mildly biasing a free-slope fit. The calibrated chain (fixed slope, §3) is robust to this at the few-percent level in distance; the free slope is reported as a diagnostic, not used in the distance.

3. The three measurement modes

The goal is always the same number — the magnitude at $B-V = 0.6$ — but real light curves come in three conditions. The pipeline chooses the mode automatically and records the choice.

Mode L — linear (the classical method)

When the linear region is well populated, fit a weighted straight line to B versus $B-V$ in the window and evaluate it at $B-V = 0.6$:

$$B(c) = B_{BV0.6} + \beta_{BV} (c - 0.6)$$

The calibrated chain fixes $\beta_{BV} = 1.94$ (the K -corrected population slope; W03) — this is what the M_{BV} calibration of §6 was constructed with. The free slope β_{free} is also fit and reported with its error as a *diagnostic*: values far from 1.94 on clean, single-source photometry indicate either peculiarity or time-variable extinction (see §7's SN 2006X note). A free-slope value is considered valid only if the fit has ≥ 5 nights, color span ≥ 0.25 mag, no internal color gap > 0.4 mag, rms ≤ 0.15 mag, and β_{free} within the WS06 validity cut $[1.5, 2.5]$.

Mode Q — quadratic interpolation

When the linear region is not obvious — sparse windows, curved (parabolic) loci, and 91bg-like objects for which W03 showed the linear model fails — $B_{BV0.6}$ is still perfectly well defined *as the magnitude where the locus crosses $B-V = 0.6$* . Mode Q fits

$$B(c) = a_0 + a_1 (c - 0.6) + a_2 (c - 0.6)^2$$

over the post-maximum branch and reports $a_0 \pm \sigma$. Interpolation only: the observed colors must bracket 0.6 (or approach it within 0.05 mag), otherwise the mode fails with a named gate (`no_bracket`). For 91bg-like objects ($\Delta m_{15} \geq 1.7$) the *measurement* is delivered but the *standardization* is flagged uncalibrated: the $M_{BV}(\Delta m_{15})$ calibration of §6 does not extend to them.

Mode R — heavily reddened objects

When extinction pushes the entire observed branch redward of $B-V = 0.6$, extrapolating the fit back to 0.6 is forbidden — extrapolation across a magnitude or more of color is precisely the systematic that corrupts naive fits. Instead (PI prescription):

1. Estimate the reddening from the peak color: $E_{\text{guess}} = B_{\text{max}} - V_{\text{max}}$ (the intrinsic peak color of a normal Ia is ≈ 0 ; optionally refine by subtracting the intrinsic color locus $\mathcal{E}_0(\Delta m_{15})$ of Eq. W03-8 below).
2. Measure the magnitude at the shifted target color $c^* = 0.6 + E_{\text{guess}}$, by Mode L or Mode Q, strictly by interpolation within the observed branch. Record (m^* , c^* , E_{guess} , β_{used}).
3. When the true reddening E_{true} is later determined (from §5, from external data, or from a dust-law analysis), apply the deferred correction:

$$\begin{aligned} B_{\text{BV}0.6} &= m^* - \beta_{\text{used}} (c^* - 0.6 - E_{\text{true}}) - 0.6 \beta_{\text{used}} - R_B E_{\text{true}} + 0.6 \beta_{\text{used}} \\ &= m^* - \beta_{\text{used}} (c^* - 0.6) + (\beta_{\text{used}} - R_B) E_{\text{true}} \end{aligned}$$

For a linear locus this is algebraically identical to the classical chain (measure at observed 0.6, correct by $(R_B - \beta)E$) — Mode R changes *where the locus is evaluated*, never the physics — but it avoids the extrapolation entirely. The host dust law R_B is configurable per object (default 3.1; e.g., SN 2006X requires its known anomalous $R_B \approx 2.5$, and Mode R makes the dust law the *only* open parameter for such objects).

4. K-corrections (the WS06 scheme)

Five raw outputs receive K-corrections: B_{max} , V_{max} , Δm_{15} , $B_{\text{BV}0.6}$, and β_{BV} . Following WS06 (their Appendix D): each correction is linear in the supernova's raw color E , with an intercept $\mu^k(z)$ and slope $\psi^k(z)$ that are cubic polynomials in redshift, calibrated from a spectrophotometric library of reddened/unreddened light-curve pairs. The package evaluates the polynomials at the object's z and raw color. At $z \leq 0.03$ the corrections are ≤ 0.01 mag; they are always applied and always quoted. (Reproduction note: the distance-relevant $K(B_{\text{BV}})$ reproduces the WS06 worked example to ~ 1 mmag; the published per-band table entries for $B_{\text{max}}/V_{\text{max}}$ include light-curve-refit effects beyond the tabulated polynomials, an ambiguity ≤ 0.01 mag in μ at these redshifts, documented in the validation suite.)

5. Extinction

Milky-Way extinction is removed first (Cardelli law, $R_B = 4.15$, with $E(B-V)_{\text{MW}}$ from the Schlafly & Finkbeiner maps). The host component uses CMAGIC's internal estimator (W03 Eqs. 7–9): the color-excess measure

$$\begin{aligned} E_{\text{cal}} &= (B_{\text{max}} - B_{\text{BV}0.6}) / \beta_{\text{BV}} && (\text{W03 Eq. 7}) \\ E_{\text{cal}_0} &= (-0.118 \pm 0.013) + (0.249 \pm 0.043)(\Delta m_{15} - 1.1) && (\text{W03 Eq. 8}) \\ E_{\text{host}} &= E_{\text{cal}} - E_{\text{cal}_0} && (\text{W03 Eq. 9}) \end{aligned}$$

with $\mathcal{E}_0$ the intrinsic (dust-free) locus of the same quantity as a function of decline rate. The host correction to the candle is $A_{\text{BV}} = (R_B - \beta_{\text{BV}}) \cdot E_{\text{host}}$ with the host-law $R_B = 3.1$ by default (override per object when the dust law is known to be anomalous). E_{host} absorbs *color calibration errors* of the photometry as if they were dust — the reason the pipeline enforces single-source photometry (§7).

6. Standardization and the distance

The absolute calibration is the W03 Table 3 linear law $M_{BV}(\Delta m_{15}) = a + b \cdot \Delta m_{15}$, constructed on color-restricted subsamples ($B_{\max} - V_{\max} \leq 0.05 / 0.2 / 0.5$) with dispersions 0.07–0.16 mag, on the $H_0 = 65$ scale (rescaled explicitly; the H_0 dependence is a single coherent $5 \log(H_0/65)$ shift). The WS06 refinement — a bilinear decline-rate law with a kink at $\Delta m_{15} = 1.1$, which beats a quadratic at 43:1 odds — is available as an option (zero point anchored to W03 Table 3 at $\Delta m_{15} = 1.1$). Then

$$\mu = B_{BV0.6}(\text{corrected}) - M_{BV}(\Delta m_{15}), \quad D = 10^{\{(\mu-25)/5\}} \text{ Mpc}.$$

Error budget per object: the fit error on $B_{BV0.6}$ (typically 0.02–0.06 mag on good archival photometry), the calibration dispersion (0.07–0.16 mag, dominant), the extinction-estimator error propagated through $(R_B - \beta)$, and the K-correction ambiguity (≤ 0.01 mag at $z \leq 0.03$).

7. Failure semantics and provenance (what makes this pipeline safe to use blind)

Every fit either returns a distance or a named gate, never a silently degraded number:

gate	meaning
window_empty	the decline-rate window contains no valid nights (e.g., 91bg-like: the color maximum arrives so early that no linear branch exists — that is the physical result)
too_few_nights	< 4 nights in the window
no_bracket	Mode Q/R interpolation impossible: the branch never crosses the target color
rms_gate / beta_gate	Mode L quality gates failed
dm15_provenance	decline rate unmeasured or from a bound-hit fit: no trustworthy window
photometry_defect	upstream data flagged (duplicated columns, cross-instrument artifacts)

Provenance rules learned from archival forensics, enforced automatically:

- One photometric source per fit (never mix observatories in a band; preference order CSP > CfA > KAIT/LOSS > other; Swift/UVOT optical bands are never used). Mixed compilations inflate the CMAGIC rms three- to five-fold and bias the internal extinction estimator.
- De-duplication: rows duplicating the same publication via different aggregators are dropped by bibcode matching.
- Every result carries: source used, nights selected, phase and color spans, mode, $\beta_{\text{free}} \pm \sigma$, rms, all gates evaluated, and the diagnostic panel (locus, window boundaries, fit, color-curve inset, annotation block).

A diagnostic worth knowing about: on clean single-source photometry, a *valid* free slope significantly different from 1.94 carries physics. SN 2006X — with documented time-variable circumstellar absorption — shows $\beta_{\text{free}} = 2.25\text{--}2.28$ (rms at calibration quality), intermediate between the intrinsic slope and its anomalous dust vector $R_B \approx 2.5$: time-variable extinction slides points along the dust vector. The free slope is CMAGIC’s built-in dust-variability alarm.

References

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 - Wang, L., Strovink, M., Conley, A., Goldhaber, G., Kowalski, M., Perlmutter, S., & Siegrist, J. 2006, ApJ, 641, 50
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Part II — CMAGIC at High Redshift

At $z \lesssim 0.1$, single-filter K-corrections suffice and Part I applies as written. At higher redshift the observer-frame filters no longer sample the rest-frame B and V bands at all: cross-filter K-corrections are required — effectively interpolations of the observed multi-band photometry, guided by a spectral template, from the observer frame to the rest-frame B and V light curves that CMAGIC needs. This part formulates that extension, its sparse-data estimator, and its validation.

8. The cross-filter problem and the template-synthesis solution

For a supernova at redshift z observed in filters $\{X_1 \dots X_n\}$ (Rubin/LSST ugrizy, Roman WFI, JWST NIRCам, or the standard UBVRIZ/ugriz/JHK sets), rest-frame B at rest phase p samples the SED near $4400(1+z)$ Å in the observer frame — between, or beyond, the observed bands. The classical per-band cross-filter K-correction (Kim, Goobar & Perlmutter 1996) transfers each observed magnitude to a chosen rest band. We adopt the equivalent but more robust template-synthesis formulation, which the pipeline implements end to end:

1. Rest-frame clock. Phases are computed as $p = (t - t_0)/(1+z)$: cosmological time dilation is applied before anything else, and the CMAGIC window of §2, the decline rate Δm_{15} , and the color-curve turnback are all defined in *rest-frame* days.
2. Template. A spectral time series $F_T(p, \lambda)$ of a normal Ia — either the Hsiao template (the simple default the method needs nothing fancier than) or the SALT3-NIR model (Pierel et al. 2022), whose wavelength coverage extends into the near-infrared and is preferred when the observed bands redshift beyond ~ 8500 Å rest (Roman and JWST at $z \gtrsim 1$, where rest-frame B–V is sampled by observer-frame YJH).
3. Warp to the data. The template is redshifted — $F_T(p, \lambda(1+z))/(1+z)$, i.e. wavelengths stretched and the SED dimmed per bandpass convention — and constrained by the observed photometry: in Hsiao mode, a stretch (from the fitted rest-frame decline) plus a smooth per-epoch color warp (a grey term and an extinction-law-shaped tilt) fitted so the template’s synthetic magnitudes reproduce every observed band at every epoch; in SALT3-NIR mode, the standard (t_0, x_0, x_1, c) fit. The warp is smooth in wavelength by construction, so it interpolates *between* the observed bands without inventing spectral features.
4. Synthesize rest-frame B, V. At each observation epoch, rest-frame B and V magnitudes are synthesized from the warped template through the Bessell B and V transmissions. The output is exactly what Part I consumes: nightly B and B–V, now in the rest frame, with the K-correction, the $(1+z)$ dilation, and the wavelength stretch all inside the synthesis.

5. Errors and covariances by Monte Carlo. The observed photometry is perturbed by its errors (default 200 realizations), the warp and synthesis are redone per realization, and the empirical covariance of the synthesized (B_i, V_i) — including the B–V covariance within an epoch and correlations between epochs induced by the shared warp — is propagated into the CMAGIC fit. A template-choice systematic (Hsiao versus SALT3-NIR difference, in the bands actually used) is reported alongside.

Filter transmissions: the package ships with (or registers on demand from `sncosmo`) the Bessell UBVRI, SDSS/LSST ugrizy, Roman WFI, and 2MASS JHK curves; any other filter — JWST NIRCам included — is supplied as a two-column (wavelength, transmission) file via `cmagic.highz.register_filter()`. The inputs per supernova are then exactly as the design requires: the light curves, the filter transmissions, and the redshift.

9. Sparse light curves: the fixed-slope estimator with full covariance

High- z light curves are often sparse: one or two epochs may land in the CMAGIC linear window. The linear fit of Mode L is then impossible — but the hypothesis of §1 does not require fitting the slope, because the slope is a population constant. Mode S (sparse): with β fixed at 1.94, every single point in the window is an estimate of the candle,

$$\begin{aligned} m_i &= B_i - \beta (c_i - 0.6), \quad c_i \equiv B_i - V_i \\ &= (1-\beta) B_i + \beta V_i + 0.6 \beta \end{aligned}$$

with variance following from the epoch’s (B, V) covariance matrix — note that B appears on both axes of the diagram, so the B–V covariance term is *not* optional:

$$\text{Var}(m_i) = (1-\beta)^2 \text{Var}(B_i) + \beta^2 \text{Var}(V_i) + 2\beta(1-\beta) \text{Cov}(B_i, V_i)$$

Multiple sparse points are combined by generalized least squares with the full covariance

$$V_{ij} = \text{Cov}(m_i, m_j) + \sigma_\beta^2 (c_i - 0.6)(c_j - 0.6) + \delta_{ij} \sigma_{\text{int}}^2$$

where the first term carries the synthesis-induced inter-epoch correlations (§8.5), the second propagates the population slope uncertainty $\sigma_\beta = 0.16$ — *fully correlated* across the epochs of one supernova, and vanishing for points at the fiducial color — and σ_{int} is the intrinsic scatter of the locus about the line (set from the low- z training sample). The slope term makes explicit what sparse data cost: a single point at $c = 0.9$ pays $0.16 \times 0.3 \approx 0.05$ mag of slope systematic; a point near $c = 0.6$ pays almost nothing — so the *placement* of the surviving epochs, not just their number, sets the error. Mode S reports `n_window`, the color leverage $|c - 0.6|$, and the slope-systematic share of the error budget, and it is gated exactly like the other modes (`window_empty` when nothing survives; no silent degradation).

10. Validation and application scope

Validation: the SDSS-II supernova survey (ugriz light curves, $z \approx 0.05$ – 0.4 , published photometry and redshifts) — deep enough that cross-filter synthesis is genuinely exercised, low enough that results can be checked against the standard-candle expectation. The validation applies the full chain blind (template synthesis → mode selection, with Mode S carrying the sparse tail) and reports the Hubble diagram against Λ CDM, the residual rms by redshift bin and by mode, the Hsiao-versus-SALT3-NIR template systematic, and the failure census with named gates. Passing criteria are honest rather than cosmetic: residual rms consistent with the low- z dispersion of §6 plus the propagated sparse-mode errors, and no mode- or z -dependent bias exceeding its error.

Application scope. With filters registered, the identical call serves Rubin (ugrizy to $z \sim 0.4$ in rest-B-V; deeper with y), Roman WFI (rest-frame B-V to $z \gtrsim 2$), and JWST NIRCам follow-up photometry. The cookbook’s rule of thumb: choose observer bands so that $4400(1+z)$ Å and $5500(1+z)$ Å each fall *inside* the covered wavelength range — synthesis interpolates well and extrapolates badly, and the pipeline gates (rest_band_uncovered) when a rest band would require extrapolating the template beyond the observed filters.

8.1 The iterative template fit, and why B_BV0.6 is stable against template details

The SALT3-NIR parameters cannot be fixed in one pass, because the K-correction and the light-curve fit depend on each other. The engine iterates (PI prescription):

1. make a first guess of (x_1, c) — from the raw observed decline and color;
2. compute the K-correction / rest-frame synthesis with the template at that (x_1, c) ;
3. fit the light-curve template to the synthesized rest-frame light curves to derive new (x_1, c) ;
4. return to step 2; exit when (x_1, c) converge (default: $|\Delta x_1| < 0.05$ and $|\Delta c| < 0.01$, typically 2–4 iterations; the pipeline gates with kcorr_no_convergence otherwise).

Convergence is benign for a reason worth stating plainly: the K-correction is essentially a color correction. Once the template — *any* reasonable Ia template — is morphed to match the observed colors, the synthesized B magnitude at the fiducial color $B-V = 0.6$ is a stable quantity, insensitive to the residual details of the template. The fiducial-color evaluation point is what buys this: template imperfections move points *along* the color–magnitude locus far more than they move the locus itself, and reading the locus at fixed color cancels the along-track freedom.

This is not an assumption but a measurable property, and the pipeline measures it: the template-insensitivity test recomputes the synthesis with templates of deliberately different (x_1, c) — and with the Hsiao template in place of SALT3-NIR — each morphed to the same observed colors, and reports the spread of the resulting B_BV0.6 as the template systematic (superseding the simpler Hsiao-versus-SALT3-NIR delta as the quoted systematic; the pass criterion is that this spread be small against the photometric error, which is the quantitative form of the stability claim above).

10.1 Worked example: the SDSS-II validation run (v0.2.0)

Blind run of `examples/sdss_validation.py` on 60 spectroscopically confirmed SDSS-II SNe Ia (z 0.05–0.35; public release of Sako et al. 2018 — see the script header for the exact source URLs), engine hsiao primary with the §8.1 insensitivity test per object:

- 36 of 60 yield distances — modes: L $\times 17$, S $\times 19$. Failure census (all named): no_bracket $\times 11$ (blue branches never reaching $B-V = 0.6$ inside the window), rms_gate $\times 9$, window_empty $\times 3$, too_few_nights $\times 1$.
- Hubble residuals about Λ CDM ($H_0 = 72$, $\Omega_m = 0.3$): grey zero-point offset -0.17 mag (the transfer of the Vega/W03 calibration through synthesis; documented in KNOWN_ISSUES, not recalibrated away), rms about the median 0.44 mag overall: mode L 0.24 mag, mode S 0.56 mag; by redshift bin 0.33 / 0.53 / 0.38 (0.05–0.15 / 0.15–0.25 / 0.25–0.35).
- Error-budget honesty: $\chi^2/\text{dof} = 3.9$ (L) and 9.4 (S) — the propagated errors underestimate the observed scatter by $\sim 2\times$ (L) and $\sim 3\times$ (S). The L-mode excess over the §6 calibration dispersion plausibly carries SMP photometry systematics and the host-extinction estimator applied to synthesized peak magnitudes; the S-mode excess concentrates at high color leverage (the §9 warning

realized: single points far from the fiducial color). The §10 passing criterion is NOT met at v0.2.0; the numbers are reported as measured.

- Template-insensitivity (§8.1): median $B_{BV0.6}$ spread over $\{x_1 \pm 0.5, c \pm 0.05, \text{Hsiao}\} = 0.056$ mag, 90th percentile 0.091 — the stability claim holds at the few-hundredths level, and per-object spreads are carried as $K_{\text{systematic}}$.